

Effective Action and Renormalization in ϕ^3 Theory

Perimeter Scholars International, Quantum Field Theory II, Tutorial 5; 7 worked parts.

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1. The Effective Action for ϕ^3 Theory

In six-dimensional Euclidean spacetime, consider

$$S_\lambda = \int d^6x \left[\frac{1}{2}(\partial\phi)^2 + \frac{m^2}{2}\phi^2 + \frac{\lambda}{3!}\phi^3 \right]. \quad (1.1)$$

For a bosonic theory, the one-loop effective action is

$$\Gamma[\phi] = S[\phi] + \frac{\hbar}{2} \text{Tr}[\ln(S''[\phi])] + O(\hbar^2), \quad (1.2)$$

where terms of order $\hbar^n$ correspond to n -loop diagrams.

- a) Calculate the integral kernel associated with the operator $S''[\phi]$ in ϕ^3 theory.

Solution.

$$S''[\phi]_{xy} = (-\partial^2 + m^2 + \lambda\phi(x)) \delta^d(x - y). \quad (1.3)$$

- b) Factor the free-field contribution out of $S''[\phi]$ and expand $\text{Tr}[\ln(S''[\phi])]$ perturbatively in λ .

Hint: Recall

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \frac{x^6}{6} + \dots. \quad (1.4)$$

Solution. Write

$$(-\partial^2 + m^2)\Delta(x - y) = \delta^d(x - y) \iff (-\partial^2 + m^2)_{xy}\Delta_{yz} = \delta_{xz}. \quad (1.5)$$

Then, we have

$$S''[\phi]_{xy} = (-\partial^2 + m^2)_{xz} (\delta_{zy} + \Delta_{zy}\lambda\phi(z)). \quad (1.6)$$

This immediately follows

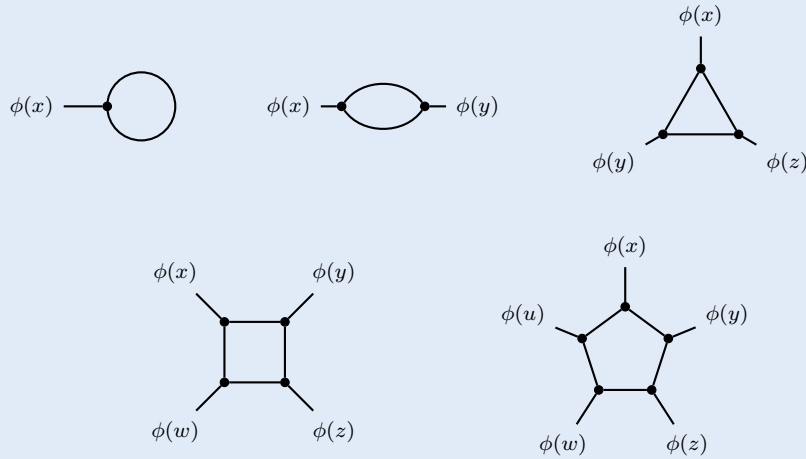
$$\begin{aligned} \text{Tr} \ln(S''[\phi]) &= \text{Tr} (\ln(-\partial^2 + m^2)_{xz} + \ln(\delta_{zy} + \lambda\Delta_{zy}\phi(z))) \\ &= \ln(\det(-\partial^2 + m^2)) + \text{Tr} \left(\lambda\Delta_{zy}\phi(y) - \frac{\lambda^2}{2}\Delta_{zu}\Delta_{uy}\phi(z)\phi(u) + \dots \right) \\ &= \ln(\det(-\partial^2 + m^2)) + \lambda\Delta(0) \int d^d x \phi(x) - \frac{\lambda^2}{2} \int d^d x d^d y \phi(x) \Delta^2(x - y) \phi(y) + \dots. \end{aligned} \quad (1.7)$$

- c) Express the complete one-loop effective action $\Gamma[\phi]$ as a power series in λ through fifth order. Represent the result using one-loop amputated Feynman diagrams.

Solution. Up to fifth order in λ , we have

$$\begin{aligned}
 \Gamma[\phi] &= \int d^d x \left[\frac{1}{2}(\partial\phi)^2 + \frac{m^2}{2}\phi^2 + \frac{\lambda}{3!}\phi^3 + \frac{\hbar\lambda}{2}\Delta(0)\phi(x) \right] \\
 &\quad - \frac{\hbar\lambda^2}{4} \int d^d x d^d y \phi(x)\Delta^2(x-y)\phi(y) \\
 &\quad + \frac{\hbar\lambda^3}{6} \int d^d x d^d y d^d z \Delta(x-y)\Delta(y-z)\Delta(z-x)\phi(x)\phi(y)\phi(z) \\
 &\quad - \frac{\hbar\lambda^4}{8} \int d^d x d^d y d^d z d^d w \Delta(x-y)\Delta(y-z)\Delta(z-w)\Delta(w-x)\phi(x)\phi(y)\phi(z)\phi(w) \\
 &\quad + \frac{\hbar\lambda^5}{10} \int d^d x d^d y d^d z d^d w d^d u \Delta(x-y)\Delta(y-z)\Delta(z-w)\Delta(w-u)\Delta(u-x) \\
 &\quad \quad \times \phi(x)\phi(y)\phi(z)\phi(w)\phi(u) \\
 &\quad + \mathcal{O}(\lambda^6) + \mathcal{O}(\hbar^2).
 \end{aligned} \tag{1.8}$$

where we have ignore overall constants. These corresponds to the diagrams



- d) Explain how the coefficients in the expansion from the preceding part are related to the symmetries of the corresponding graphs.
- e) Use the one-loop effective action to show that quantum effects in ϕ^3 theory generate a vacuum expectation value for $\phi(x)$.

Hint: The quantum vacuum field $\varphi(x)$ is the configuration that extremizes the quantum effective action.

Solution. Since there is now a term of order 1 in ϕ , there is a constant term in the effective equation of motion, $\Gamma'[\varphi] = 0$. Therefore, φ must be nonzero, and by the definition of φ , the VEV of ϕ^3 theory is nonzero.

2. Renormalization

- a) Redefine the field so that it creates a normalized one-particle state with no vacuum overlap:

$$\langle \Omega | \phi(0) | \Omega \rangle = 0, \quad \langle \mathbf{k} | \phi(0) | \Omega \rangle = 1. \tag{2.1}$$

Show that the Lagrangian can then be written as

$$\mathcal{L} = \frac{1}{2} A \partial_\mu \phi \partial^\mu \phi + B \phi + \frac{1}{2} C \phi^2 + \frac{1}{3!} D \phi^3. \tag{2.2}$$

Solution. In general, the bare field would give

$$\langle \Omega | \phi_0(0) | \Omega \rangle = v, \quad \langle \mathbf{k} | \phi_0(0) | \Omega \rangle = \sqrt{Z}. \quad (2.3)$$

Define the renormalised field as

$$\phi(x) = \frac{1}{\sqrt{Z}} \phi_0(x) - v, \quad (2.4)$$

so that

$$\langle \Omega | \phi(0) | \Omega \rangle = 0, \quad \langle \mathbf{k} | \phi(0) | \Omega \rangle = 1. \quad (2.5)$$

Plugging this into the Lagrangian, we find

$$\mathcal{L} = \frac{1}{2} A \partial_\mu \phi \partial^\mu \phi + B \phi + \frac{1}{2} C \phi^2 + \frac{1}{3!} D \phi^3 \quad (2.6)$$

after some redefinition.

- b) The factor A ensures the equal-time canonical commutator $[\phi(\mathbf{x}), \Pi(\mathbf{y})] = i\hbar\delta(\mathbf{x} - \mathbf{y})$. Use the six-dimensional Källén–Lehmann representation

$$\langle \Omega | \phi(x) \phi(y) | \Omega \rangle \quad (2.7)$$

$$= \int \frac{d^5 \mathbf{k}}{(2\pi)^5} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 + \int_{4m^2}^{\infty} ds \int \frac{d^5 \mathbf{k}}{(2\pi)^5} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} \rho(s) \quad (2.8)$$

to find the relation between A and $\rho(s)$.

Solution. Consider

$$\begin{aligned} \partial_{y_0} \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle \Big|_{x_0=y_0} &= \frac{i}{2} \int \frac{d^5 \mathbf{k}}{(2\pi)^5} (e^{-i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})} + e^{i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})}) \\ &\quad + \frac{i}{2} \int_{4m^2}^{\infty} ds \int \frac{d^5 \mathbf{k}}{(2\pi)^5} (e^{-i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})} + e^{i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})}) \rho(s) \\ &= i\delta^5(\mathbf{x} - \mathbf{y}) + i \int_{4m^2}^{\infty} ds \rho(s) \delta^5(\mathbf{x} - \mathbf{y}). \end{aligned} \quad (2.9)$$

Meanwhile, we have

$$\Pi(x) = Z \partial_0 \phi(x). \quad (2.10)$$

Therefore, we have

$$\partial_{y_0} \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle \Big|_{x_0=y_0} = \frac{1}{Z} \partial_{y_0} \langle \Omega | [\phi(\mathbf{x}), \Pi(\mathbf{y})] | \Omega \rangle \implies 1 = Z \left(1 + \int_{4m^2}^{\infty} ds \rho(s) \right). \quad (2.11)$$

- c) Impose four renormalization conditions to determine A, B, C, D . Two are the normalization conditions above; a third sets the physical mass to zero. For the fourth, require

$$\Gamma(p_1, p_2, p_3) = \lambda_R \quad \text{when} \quad (p_1 + p_2)^2 = (p_1 + p_3)^2 = \mu^2. \quad (2.12)$$

With a sharp momentum cutoff Λ , determine A, B, C in terms of λ_R and Λ to the order requested, and determine D through order λ_R^2 .

Solution. First, let us rewrite the Lagrangian as

$$\mathcal{L} = \frac{1}{2} (\partial\phi)^2 + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{3!} \phi^3 + \frac{\delta_A}{2} (\partial\phi)^2 + B\phi + \frac{1}{2} \delta_C \phi^2 + \frac{\delta_D}{3!} \phi^3, \quad (2.13)$$

where $\delta_A = A - 1$, $\delta_C = C - m^2$, and $\delta_D = D - \lambda$. Let us start with B , at one loop order, the

only contributing diagram is

$$\begin{aligned}
 & \text{---} \bullet \text{---} \bigcirc \text{---} + \text{---} \times \text{---} \\
 &= -\frac{\lambda}{2} \int \frac{d^6 k}{(2\pi)^6} \frac{1}{k^2 + m^2} - B.
 \end{aligned} \tag{2.14}$$

Since the vacuum expectation value must vanishes, the above must vanishes, and hence we have

$$B = -\frac{\lambda}{128\pi^3} \left[\frac{\Lambda^4}{4} - \frac{m^2 \Lambda^2}{2} + \frac{m^4}{2} \ln \left(1 + \frac{\Lambda^2}{m^2} \right) \right] \tag{2.15}$$

at one-loop order.

Now, we turn to the mass renormalisation factor. At 1-loop order, we have

$$\begin{aligned}
 -M_2(p^2) &= \text{---} \bullet \text{---} \bigcirc \text{---} \bullet \text{---} + \text{---} \times \text{---} \\
 &= \frac{(-\lambda)^2}{2} \int \frac{d^6 k}{(2\pi)^6} \frac{1}{(k^2 + m^2)((k+p)^2 + m^2)} - \delta_C - \delta_A p^2.
 \end{aligned} \tag{2.16}$$

The renormalisation condition requires

$$m^2 + M_2(0) = 0, \quad \lim_{p^2 \rightarrow 0} \frac{p^2}{p^2 + m^2 + M_2(p^2)} = 1 \implies M_2(0) = -m^2, \quad M_2'(0) = 0. \tag{2.17}$$

Therefore, we have

$$C = \frac{\lambda^2}{2} I_2(0) = \frac{\lambda^2}{256\pi^3} \left[\Lambda^2 + \frac{m^2 \Lambda^2}{\Lambda^2 + m^2} - 2m^2 \ln \left(1 + \frac{\Lambda^2}{m^2} \right) \right], \tag{2.18}$$

$$A = 1 + \frac{\lambda^2}{2} I'(0) = 1 - \frac{\lambda^2}{768\pi^3} \left[\ln \left(1 + \frac{\Lambda^2}{m^2} \right) + \frac{2m^2}{\Lambda^2 + m^2} - \frac{m^4}{2(\Lambda^2 + m^2)^2} - \frac{3}{2} \right]. \tag{2.19}$$

Since the first loop correction to the three-point function is of order λ^3 . At order λ^2 , we merely have $D = \lambda_R$.